Practice Problem Solutions: Day 32 - Unsupervised Machine Learning (Part-2) (Hinglish)

Yeh document `README_HINGLISH.md` mein diye gaye 3 case studies ke step-by-step mathematical aur logical solutions provide karta hai.

Solution to Case Study 1: E-Commerce Analytics & PCA Compression

1. Curse of Dimensionality & Hughes Phenomenon Ka Analysis

- **Euclidean Distance Ka Failsafe**: 50-dimensional space mein do customers ke beech Euclidean distance calculate karne par:

$$d(\mathbf{p}, \mathbf{q}) = \sqrt{\sum_{j=1}^{50} (p_j - q_j)^2}$$

Jab dimensions bohot badh jaate hain ($d \to \infty$):

$$\lim_{d \to \infty} \frac{\text{dist}_{\text{max}} - \text{dist}_{\text{min}}}{\text{dist}_{\text{min}}} = 0$$

Iska matlab hai ki 50 dimensions mein sabse paas wale customer aur sabse dur wale customer ka distance lagbhag barabar ho jaata hai. k-NN search mein contrast khatam ho jaata hai aur model randomly neighbors uthane lagta hai.

- **Hughes Phenomenon (Peaking Effect)**:
  - $10 \to 20$ features: Useful behavioral signals add hue, isliye recommender ki accuracy improve hui.
  - $20 \to 50$ features: Extra features sirf meaningless noise aur correlation le aaye (jaise button hover, micro-scrolls). In noisy features ne space ko exponentially bada kar diya, jisse real customer patterns noise ke neeche dab gaye.

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2. Variance Retention Calculation

Given:

- Saare 50 eigenvalues ka sum: $\sum_{j=1}^{50} \lambda_j = 40.0$
- $\lambda_1 = 15.2$
- $\lambda_2 = 8.8$
- $\lambda_3 = 5.2$

Individual Explained Variance Ratios:

$$\text{Ratio}(PC1) = \frac{\lambda_1}{\sum \lambda_j} = \frac{15.2}{40.0} = 0.3800 \implies \mathbf{38.00\%}$$

$$\text{Ratio}(PC2) = \frac{\lambda_2}{\sum \lambda_j} = \frac{8.8}{40.0} = 0.2200 \implies \mathbf{22.00\%}$$

$$\text{Ratio}(PC3) = \frac{\lambda_3}{\sum \lambda_j} = \frac{5.2}{40.0} = 0.1300 \implies \mathbf{13.00\%}$$

Cumulative Explained Variance Ratio:

$$\text{Total Variance} = 38.00\% + 22.00\% + 13.00\% = \mathbf{73.00\%}$$

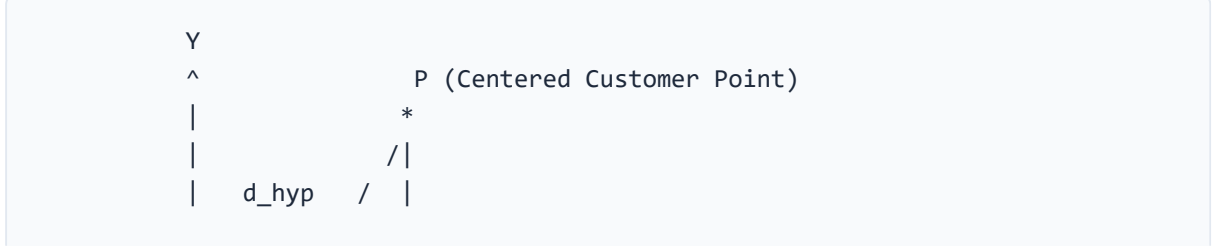
Decision:

$$\mathbf{73.00\%} \geq \mathbf{70.00\%}$$

Verdict: 50 features ko reduce karke sirf 3 components ($PC1, PC2, PC3$) rakhne se platform ki **70% requirement poori ho jaati hai**. 94% dimensions kam ho gaye, recommendation engine fast ho gaya aur 73% meaningful information bilkul safe hai!

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3. Pythagoras Theorem Ka Derivation Proof (Ma'am Ka Blackboard Method)



```
|         /    | d_1 (Perpendicular Error)
|       /      |
|     /        |
|   /          v
|  *-----------P'-----------> PC1 Candidate Line
| O     d_2 (Origin se Projection ka distance)
+-----------------------------------> X
```

1. Maan lijiye $O = (0, 0)$ origin hai.
2. P ek customer ka data point hai jiske coordinates (x_i, y_i) fix hain.
3. P' point P ka candidate line par perpendicular projection hai.
4. Triangle $\triangle OP'P$ ek right-angled triangle hai. Pythagoras theorem lagane par:

$$d_1^2 + d_2^2 = d_{\text{hyp}}^2$$

5. Kisi bhi point P ke liye origin se uska distance $d_{\text{hyp}}^2 = x_i^2 + y_i^2$ **hamesha fixed (constant)** hota hai:

$$d_1^2 = d_{\text{hyp}}^2 - d_2^2$$

6. Saare n points ke liye sum lene par:

$$\sum_{i=1}^{n} d_{1,i}^2 = \sum_{i=1}^{n} d_{\text{hyp},i}^2 - \sum_{i=1}^{n} d_{2,i}^2$$

7. Kyunki $\sum d_{\text{hyp}}^2$ ek fixed number hai:

$$\min \sum_{i=1}^{n} d_{1,i}^2 \iff \max \sum_{i=1}^{n} d_{2,i}^2 \quad \blacksquare$$

Result: Perpendicular error d_1 ko kam se kam karna, origin se projected distance d_2 ko zyada se zyada karne ke bilkul barabar hai!

Solution to Case Study 2: Silicon Wafer Defect Detection with Non-Linear DBSCAN

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1. K-Means Kyun Fail Hua?

1. **Zabardasti Cluster Assignment**: K-Means mein noise ka koi option nahi hota. Har dhool ke kan ko zabardasti kisi circuit cluster mein daal diya jaata hai.
2. **Centroid Ka Khinch Jaana**: Centroid saare points ka mean hota hai. Door kone mein gire dust particle ki wajah se centroid khinch kar empty space mein chala jaata hai, jisse actual circuit path corrupt ho jaata hai.
3. **Linear Separation**: K-Means mudi hui crescent (aadhe chand) shapes ko follow nahi kar sakta; woh straight cuts lagata hai.

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2. Rule of Thumb Se Hyperparameter Selection

- `min_samples` **Ka Rule of Thumb**:

> $$\text{min_samples} \approx 2 \times \text{features} = 2 \times 2 = 4 \implies \mathbf{4} \text{ ya } \mathbf{5} \text{ chunein}$$

- $\text{min_samples} = 100$ **Rakhne Ka Asar**: 500 points ke data mein 100 points ka bheed maangne par valid circuits bhi fail ho jaayenge aur poora data **Noise (** `-1` **)** ban jaayega.
- $\text{min_samples} = 1$ **Rakhne Ka Asar**: Ek akela dhool ka kan bhi apna alag valid cluster ghoshit ho jaayega.

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3. Epsilon Progression & Paas-Paas Ke Dust Particles

- $\epsilon = 0.50$: Radius itna bada hai ki dono alag-alag crescent lines ek hi massive cluster ban jaati hain, aur zero noise detect hota hai.
- $\epsilon = 0.18$: Dono crescent tracks alag-alag cluster 0 aur 1 bante hain, aur baaki contaminants noise mark ho jaate hain.
- **Paas Gire Do Dust Particles (**$d < \epsilon$**)**: Dono ke paas hone ke bawajood:
    1. Neighborhood mein sirf 2 points hain ($2 < \text{min_samples} = 5$), isliye **koi bhi Core Point nahi ban sakta**.
    2. Dono particles wafer ke kone mein hain, isliye kisi bhi main circuit ke Core Point se connected nahi hain.
    3. Jo point Core Point nahi hai aur kisi Core Point se connected bhi nahi hai, woh **Noise (** `label = -1` **)** hota hai. Isliye dono particles successfully defect mark honge!

Solution to Case Study 3: Clinical Thyroid Pathology: Isolation Forest vs. LOF

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1. Isolation Forest Anomaly Scoring

Anomaly Score ka official formula:

$$s(x, n) = 2^{-\frac{E(h(x))}{c(n)}}$$

Yahan $E(h(x))$ average path length hai aur $c(n) \approx 16.52$ expected BST search depth hai.

Patient $P_{\text{pathological}}$ Ka Score:

Given: $E(h(x)) = 2.10$ aur $c(6916) = 16.52$:

$$\text{Ratio} = \frac{2.10}{16.52} \approx 0.127118$$

$$s = 2^{-0.127118} \approx \mathbf{0.9156} \quad (91.56\%)$$

Classification Verdict:

- Score $s \approx 0.92 \rightarrow 1.0$ baseline 0.5 se bohot bada hai.
- **Verdict**: **Severe Hypothyroid Disease Anomaly**.
- **Scikit-Learn Label**: `-1`.

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2. 21 Dimensions Mein Speed Ka Fayda

Isolation Forest $O(t \cdot \psi \log \psi)$ linear time mein execute hota hai. Isko 6,916 points ke beech koi pairwise Euclidean distance matrix ($O(N^2 d)$) calculate nahi karni padti. Random cuts seedhe individual features par lagte hain, isliye 21 dimensions mein bhi yeh superfast aur accurate rehta hai.

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3. Demographic Subgroups & LOF Solution

- **Global Method Ka Failure**: Agar bujurg mareezon ka normal TSH $1.5 - 2.5$ hai aur young mareezon ka normal TSH $4.0 - 5.5$ hai, toh kisi 20 saal ke athlete ka TSH gir kar 2.2 ho jaana ek severe bimari ka sanket hai. Par global method 2.2 ko normal maan lega kyunki elderly cluster $1.5 - 2.5$ ke beech baitha hai!

- **Patient q Ka LOF Calculation**: Given: $\text{lrd}_{20}(q) = 0.05$ aur neighbors ki average density $= 0.20$:

$$\text{LOF}_{20}(q) = \frac{\text{Neighbors Ki Average Density}}{\text{lrd}_{20}(q)} = \frac{0.20}{0.05} = \mathbf{4.00}$$

- **Interpretation**:
  - Patient q ki local density uske 20 demographic peers se **4 guna kam (sparse)** hai!
  - Kyunki $\text{LOF} = 4.00 \gg 1.0$, patient q apne subgroup se isolated hai.
  - **Scikit-Learn Label**: `-1` **(Definite Local Anomaly)**.